

HEATHER M. (CORMIER) BHOWMICK

Lawrence, MA 01840 | (603) 401-0522 | heather.m.cormier@gmail.com | heatherbhowmick-portfolio.herokuapp.com

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, HTML5, CSS3, SQL

Technologies/Development Frameworks: NumPy, Pandas, Flask, jQuery, AJAX, JSON, BootstrapUI, ReactJS

Databases: SQL, MySQL, SQLite, PostgreSQL

Tools/Methodologies: ArcGIS/Pro, ArcGIS Extensions: Spatial Analyst/3D Analyst/Network Analyst, AGOL, ArcGIS API for JavaScript, Postman, CityEngine, GeoPlanner, IDRISI/TerrSet, ENVI, CLASlite, GeoDa, Carto, Git, GitHub, Heroku, Anaconda, Jupyter Notebook, Atom, Microsoft Office (Word, Excel, PowerPoint), MS Visio, Adobe Dreamweaver, OneDrive

PROFESSIONAL EXPERIENCE

Geographic Information Systems Consultant

Sep 2018 – Present

Southfield Redevelopment Authority, South Weymouth, MA

- Consolidated and managed datasets; created new shapefiles from CAD drawings and through georeferencing paper maps.
- Created ArcGIS Online web maps and applications to visualize historic use, phased land transfers, and redevelopment plans.
- Created interactive, dynamic planning model to engage developers, stakeholders, and public.

Software Developer

Jan 2019 – Mar 2019

Coding Temple, Boston, MA

- Built backend applications using Python, Flask frameworks.
- Exported/Imported data between different data sources using SQL.
- Used Python NumPy, SciPy, Pandas packages to perform dataset manipulation.
- Composed complex SQL queries to extract data from relational databases.

Geographic Information Systems Analyst

Nov 2015 – Aug 2018

Cape Cod Commission, Barnstable, MA

- Created, edited, updated, and managed new and existing datasets, while maintaining appropriate and detailed metadata.
- Mapped bicycle and railroad routes and interpolated elevation for profiles used in transportation planning decision-making.
- Developed methodology, using Network Analyst, to assess pedestrian access to popular community destinations across Cape Cod; conducted gap analysis by comparing existing to ideal conditions and prioritized needs for improved infrastructure.
- Classified raster imagery into land covers using supervised/unsupervised classification, segmentation, and feature extraction.
- Worked on variety of ongoing, long-term projects, such as identifying regional activity centers across Cape Cod to update county's Regional Policy Plan documents and inform new regulations; documented methods in ArcGIS Online story map.
- Created statistical benchmarks of economy, housing, water, etc., by integrating assessor/census/agency datasets; worked with Web Development team to design and build interactive web application to generate report cards for user-defined geographies.
- Conducted spatial fiscal impact analysis to allocate town revenues and expenditures to individual parcels, based on geographically distributed factors; estimated fiscal impacts of different land uses, as well as new and redevelopment projects.
- Designed and delivered presentations of project methods/results and offered recommendations to internal and external teams.

Adjunct Instructor/Lecturer – Introduction to Geographic Information Systems

Mar – May 2016

Worcester Polytechnic Institute, Worcester, MA

- Prepared content and lectured twice per week for a class of twenty-five undergraduate students, covering topics including software basics, editing, data management, databases, cartography, geocoding, raster analysis, and coordinate systems.
- Adapted course syllabus, twelve supplemental assignments, and final project to meet diverse interests and goals.
- Maintained consistent office hours and thoughtfully and efficiently graded all written and oral assignments.

Geographic Information Systems Intern

Jun – Sep 2015

Cape Cod Commission, Barnstable, MA

- Analyzed forest loss and development gain across Cape Cod (2001- 2011), using National Land Cover Database data.
- Developed land use pattern metrics to compare Cape Cod to U.S. counties with similar seasonal tourism character, as part of Barnstable Area Regional Trends initiative and to evaluate success of the Cape Cod Commission's regulatory efforts.
- Extracted geospatial information by town to calculate and compare socioeconomic indicators.

Teaching Assistant – Advanced Raster Geographic Information Systems

Jan – May 2015

Clark University, Worcester, MA

- Attended lectures and graded weekly tutorial assignments, as well as exams and projects.
- Held weekly office hours; helped fifty students to master concepts/skills and plan/execute final projects.

Geographic Information Systems Intern

May – Jul 2014

Town of Andover, Andover, MA

- Georeferenced and digitized conservation easements from survey maps and conducted census data analyses.
- Collected GPS locations and designed map templates with standardized symbols for recreation project.
- Attended project scope meetings with members of the planning and engineering divisions.
- Updated common database and created work instructions detailing steps for replication.

EDUCATION

Master of Science, Geographic Information Science for Development and Environment Clark University , Worcester, Massachusetts Graduated Summa Cum Laude, 4.23 of 4.3 G.P.A.	May 2015
Bachelor of Science, Economics-Finance, Quantitative Perspectives Bentley University , Waltham, Massachusetts Minor: Information Design and Corporate Communication Graduated Summa Cum Laude, 3.94 of 4.0 G.P.A.	May 2010
Study Abroad Program Vesalius College , Brussels, Belgium	Jan – May 2008

OTHER EXPERIENCE

GISDE Computer Lab Manager, Clark University, Worcester, MA	Jan – Dec 2014
Contract Development & Assessment Specialist/Analyst	May 2012 – Jul 2013
Contract Operations Specialist	Jul 2010 – May 2012
Sunovion Pharmaceuticals, Inc., Marlborough, MA • Paid Medicaid/Medicare rebate invoices while maintaining Sarbanes-Oxley documentation for thorough record keeping. • Analyzed and disputed individual Medicaid claims, resulting in over \$800,000 saved for the company. • Composed and executed new agreements/amendments for twenty-four regional pharmacy benefit accounts. • Analyzed financial models and examined net sales forecasts for strategic product formulary positioning. • Managed team's contract repository; developed tracking log to organize contract updates and highlight critical deadlines.	
Finance Specialist Intern, InVivo Therapeutics Corporation, Cambridge, MA	Jun – Jul 2009
Outside Plant Engineering and Planning Intern, FairPoint Communications, Manchester, NH	Dec 2008 – Jan 2009
Business Intern, Lexington Insurance Co., Boston, MA	Jun – Aug 2008
Outside Plant Engineering and Planning Intern, Verizon New England Inc., Manchester, NH	May – Aug 2007

PROFESSIONAL PRESENTATIONS & PROJECTS

<i>Python and Data Science Capstone</i>	Apr 2019
Presented (Web Application) at Coding Temple Showcase (Boston, MA) • Built program to map and visualize urban redevelopment plans in relation to existing character. • Integrated Flask framework to consume form requests and handle geospatial data analysis functions written in Python. • Used ArcGIS API for JavaScript to render and toggle between both 2D and 3D views, as well as edit and update hosted feature layers on ArcGIS Online data portal.	
<i>Mapping for a Regional Growth Policy</i>	Oct 2016
Presented (Technical Session) at Fall Northeast Arc User Group Conference (Falmouth, MA) • Analyzed community characteristics, including civic activity, commercial activity, and physical form to define diverse built environments across the fifteen towns making up the region of Cape Cod. • Delineated boundaries for sixteen existing centers of activity to focus future growth and/or redevelopment strategies. • Published story map journal application, which consumes custom web maps with uniquely configured pop-ups.	
<i>Establishing a Plant Communities Inventory</i>	Oct 2016
Presented (Poster) at Fall Northeast Arc User Group Conference (Falmouth, MA) • Combined high-resolution aerial photography and LiDAR-derived data to classify plant species within MA Route 6 median. • Evaluated overall density and health of vegetation for future stormwater and landscape management planning.	
<i>Landscape Assessment of White-Lipped Peccary and King Vulture Habitat in the Calakmul Biosphere Reserve in the Yucatán Peninsula, Mexico</i>	Jul 2015
Presented (Poster) at International Association for Landscape Ecology World Congress (Portland, OR) • Performed habitat suitability analysis for the white-lipped peccary and king vulture using Mahalanobis Typicality species distribution modeling. • Determined potential optimal corridors for the white-lipped peccary through a dynamic patch-corridor model. • Evaluated current conservation status of the Calakmul Biosphere Reserve and identified vulnerable areas within the reserve, representing anthropogenic mortality risk the white-lipped peccary faces due to hunting when ranging nearby.	
<i>Change in the Extent of Baffin Island's Penny Ice Cap in Response to Regional Warming, 1985 – 2010</i>	Dec 2014
Presented (Poster) at American Geophysical Union Fall Meeting (San Francisco, CA) • Digitized outlines of the spatial extent of the Penny Ice Cap and quantified total area and glacier length change and rates of change over three decades, using historical glacier atlas inventories and remotely-sensed imagery. • Conducted an accuracy assessment of Landsat-derived outlines, using finer spatial resolution SPOT imagery from 2009. • Examined total and localized spatial patterns of change in comparison to temperature and precipitation records. • Analyzed hypsometry to investigate regional elevation influences on change.	

HONORS & AWARDS

IDCE Student Association Travel Award , Clark University	2014 – 2015
Graduate Student Council Travel Award , Clark University	2014
IDCE Tuition Fellowship , Clark University	2013 – 2015
Encore Award , Sunovion Pharmaceuticals, Inc.	2011
Service-Learning Certificate , Bentley University	2010
President's List , Bentley University	2006 – 2010
Trustee Scholarship , Bentley University	2006 – 2010
Women's Association of Verizon Employees Scholarship , Verizon New England Inc.	2006 – 2007

COMMUNITY INVOLVEMENT

• The Boston Python User Group & Coding Temple – Event Assistant (Cambridge, MA)	2019
• Cape Wildlife Center – Wildlife Support Volunteer (Barnstable, MA)	2016
• American Conservation Experience (Catalina Island, CA)	2014
• Salvation Army (Worcester, MA)	2014
• Habitat for Humanity (Bedford, Lawrence, and Sturbridge, MA; South Palm Beach, FL)	2007 – 2012